2011, the Taylor rule model advocated interest rates to be much higher than what they were (interest rates were effectively zero).

Because the Fed has such a big impact on bond prices and because the Taylor rule tells us (for the most part) what the Fed will do, bond investors should pay attention to the Taylor rule as it helps them get their arms around the impact of inflation and economic growth risks. The Fed also sets short-term interest rates to be higher or lower than predicted by a pure response to only macro variables, and this is monetary policy risk. Monetary policy risk is important for bond prices: Ang, Dong, and Piazzesi (2007) estimate that monetary policy risk accounts for 25% to 35% of the variance of yield levels.

2.4. CHANGING POLICY STANCES

The basic Taylor rule framework assumes that the Fed's *reaction function* is constant—that is, the way the Fed responds to a given output or inflation shock does not vary over time, although obviously the shocks vary in size. But what if the *stance* of the Fed toward output or inflation risk changes over time? In equation (9.1), this is captured by time-varying coefficients on growth and inflation, rather than constant coefficients of 0.5 and 1.5, respectively, in the benchmark Taylor (1993) model. In addition to potentially changing Fed stances with respect to macro risk, which of the Fed's dueling mandates—full employment or stable prices—has been given precedence by the Fed?

These are the questions that Ang et al. (2011) investigate. They find that the Fed policy stances—the policy coefficients on output and inflation in the Taylor rule—have evolved a great deal over time, mostly with respect to inflation rather than output. These changing policy stances also have a big impact on long-term bond prices.

Figure 9.5 estimated by Ang et al. (2011) shows that the monetary policy loading on output has not moved very much from around 0.4, whereas the inflation loading has changed substantially from close to zero in 2003 to approximately 2.4 in 1983. Figure 9.5 shows that during the 1960s and 1970s, the response to inflation was less than 1.0. According to the Taylor principle, the coefficient on inflation should be greater than 1.0, otherwise *multiple equilibria* occur. This is economist talk for very volatile and unfortunate macro outcomes—like the high inflation and low economic growth (*stagflation*) of the 1970s.[11] The Fed became much more aggressive in battling inflation during the 1980s, when Volcker brought the inflation beast under control. In the early 2000s, the inflation loading was again very low but rose quickly in the mid-2000s.

[11] This view is argued by Taylor (1999) and Clarida, Gali, and Gertler (2000), among others. For a theory view of how multiple equilibria can arise and why this is bad, see Benhabib, Schmitt-Grohe, and Uribe (2001).